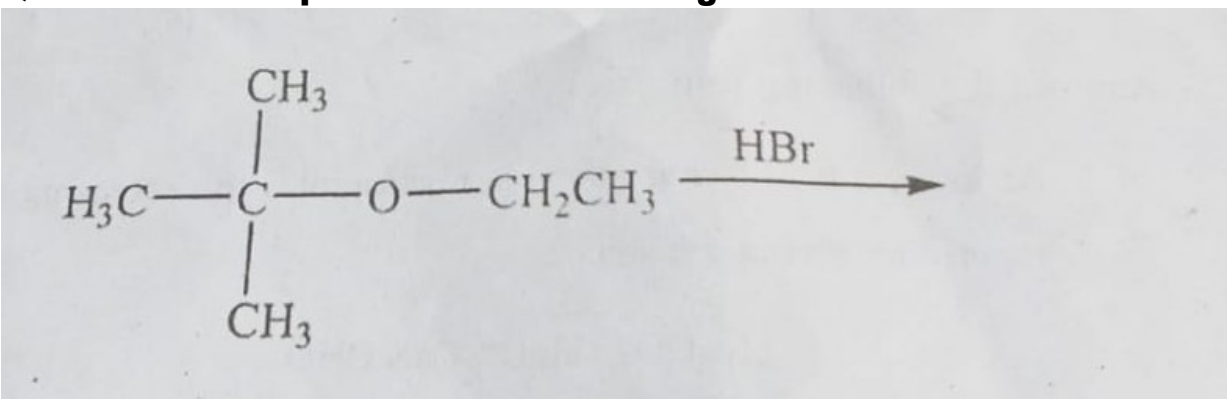
Q.No.1How many mononitration products would be formed from each isomeric dibromobenzene?
Q.No.2.Comment on acidic properties of isomeric nitrophenols.
Q.No.3. Why do aryl halides and vinyl halides show low reactivity towards nucleophilic substitution reactions compared to alkyl halides ?
Q.No.4.How many molecules of HIO4 would be consumed for oxidation of propane-1, 2, 3-triol and what are the products formed ?
Q.No.5.Giving reasons, arrange the following in increasing order of reactivity towards alkaline hydrolysis by SN_1 reaction:

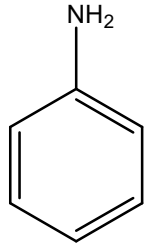
CH_3Cl, Ph_2CHCl, Ph_3CCl, $PhCH_2Cl$

Q.No.6.Write the product of the following reaction with mechanism:



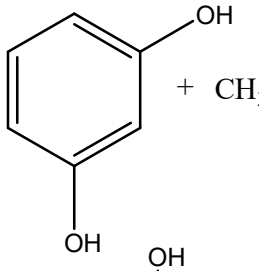
Q.No.7.Why 2, 4, 6-trinitrochlorobenzene is easily hydrolysed in the presence of aq. NaOH solution but not chlorobenzene ?
Q. No.8.The reaction of $(CH_3)_3$-C-Cl with CH_3O^- does not give of $(CH_3)_3C$-OCH_3
Q.No.8.Arrange the following alkoxide nucleophiles in decreasing order of SN_2 reactivity giving reasons:
Me_3CO^-, MeO^-, Me_2CHO^-.
Q.No.9.Why is nitrobenzene a suitable solvent for the Friedel-Crafts alkylation of PhBr while benzene is not ?
Q.No.10.Which is a stronger base, RO^- or RS^-? Which will be a better nucleophile in aqueous solution and DMSO ?
Q.No.11.Write short notes on :
(i) Oppenauer oxidation
(ii) Claisen rearrangement
(iii) Pinacol Pinacolone rearrangement
(iv) Friedel-Crafts reaction.

Q.No.12. Complete the Following Reactions:

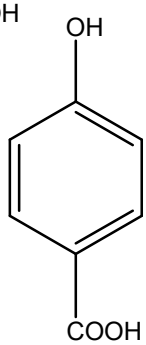
1. 

+ CH_3COCl —NaOH→ ?

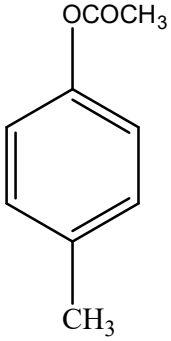
2, $CH_3CHOHCH_2CH_2CH_3$ —PCC→ ? ?

3 

+ CH_3CN —$ZnCl_2$, HCl→ ? —H_2O→ ?

4 

—Br_2→ ?

5 

—$AlCl_3$→ ?

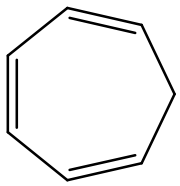
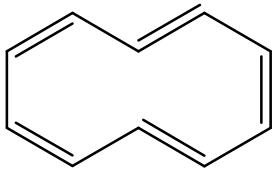
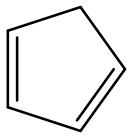
Q.No.13Give its mechanism with stereochemical aspect. What happens when tertiary butyl bromide is treated with water ?

Q.No.14.How would you distinguish between the following by a chemical reaction:

A) 1-butanol and diethyl ether

B)tert-butyl alcohol and methanol.

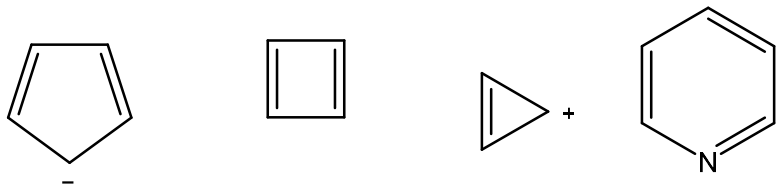
Q.No.15.Classify the following compounds as aromatic, antiaromatic and non-aromatic?

Q.No.16.Outline the following conversions
(i) Benzyl alcohol from methanal
(ii) Salicylic acid from phenol
(iii) Acetone from methyl bromide
(iv) Benzene to m-bromobenzoic acid.
(b) Both o-bromoanisole and m-bromoanisole give the same product on reaction with $NaNH_2$ in the presence of liquid ammonia.

Q.No.17.Write the product with complete chemical Reactions for the following. Name the reaction:
a) Phenol is treated with carbon dioxide under pressure in presence of sodium hydroxide.
b) Benzene is treated with isopropyl chloride in presence of $AlCl_3$,.
c) Allyl Phenyl ether is heated at temperature of 200°C.
d) Resorcinol is treated with CH_3CN in presence of $ZnCl_2$,.
e) 2-Bromopentane is treated with sodium methoxide in ethanol

Q. No.18.Give reason for the following (any three):
(a) p-Nitrophenol is more acidic than o-Nitrophenol and p-Methoxy phenol.
(b) SN_1 reactions are accompanied by racemization as well as inversion of configuration.
(c) Reactivity of aryl halides towards nucleophilic substitution increases with substitution of nitro groups at ortho and para position.
(d) Neopentyl halides are slow towards both S_N1 and S_N2 type of reactions. (543-15)
Q.No.19.
(a) Define aromaticity. Classify these compounds as aromatic or antiaromatic. Justify your answers .



R. No.20.Explain the formation of different products formed when ethyl bromide is treated with NaCN and AgCN.
Q. No.21. CH_3 group of toluene is ortho and para directing whereas CHO in benzaldehyde is meta directing in electrophilic aromatic substitution reactions" .

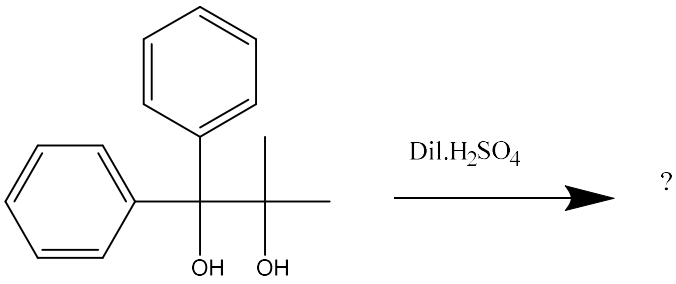
R. No.22. Illustrate the following:
(a) Internal Nucleophilic Substitution reaction with stereochemical aspects.

b) Hydrolysis of substituted epoxide under acidic and basic conditions.
(c) Industrial preparation of phenol from benzene. Why thìs process is considered to be an economical process?

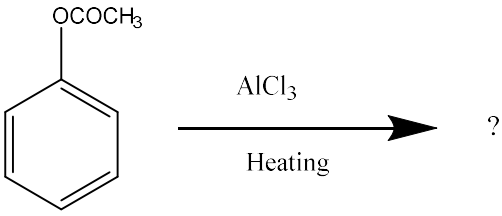
Q. No.23.Explain the following:
(a) Advantages of Friedal Craft's Acylation over Friedal Craft's Alkylation.
(b) Only m-methoxyaniline is obtained when o-bromomethoxybenzene Or bromomethoxybenzene is treated with $NaNH_2$ in the liquid NH_3
(c) Secondary and tertiary alkyl halide are not used in Willamson synthesis reaction.

Q.24.Give the mechanism for the following reactions:

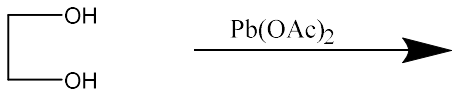
1.



2.



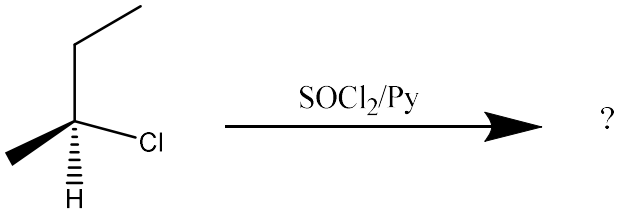
3.



4. CH3COOH + $CH_3O^{18}H$ $\xrightleftharpoons{H^+/H_2O}$?

Oxygen-18 Labelled methyl alcohol

Q.No.25.Complete Following reactions:

1 $CO_2 + CH_3CH_2MgBr \longrightarrow$? $\xrightarrow{H^+}$?

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